Xinhui CHEN

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EDUCATION

South China University of Technology Guangzhou, CN
M.Eng. of Safety Engineering *Sep. 2022 – Present*

- **Average Score**: 87 / 100
- **Core Courses**: *Case Studies on Production Accidents (91), Applied Statistics Theory and Method (93), Safety Production Technology (90), Fire Science (85)*

Northeastern University Shenyang, CN
B.Eng. of Safety Engineering *Sep. 2018 – Jun. 2022*

- **GPA**: 3.8 / 5.0 **Ranking**: 6 / 56 (Top 10.7%)
- **Core Courses**: *Mechanical Safety (97), Safety Measurement and Instrument (96), Architecture Safety (94), Emergency Technique (90), Safety Ergonomics (90)*
- **Course design**: *Fire and Explosion Prevention (A), Electric Safety (A), Accident Investigation (A)*

RESEARCH INTERESTS

- CO_2 Sequestration
- Unconventional Oil and Gas Extraction
- Computational Fluid Dynamics (CFD)
- Artificial Intelligence and Machine Learning
- Simulation, Modelling and Numerical Methods

PUBLICATIONS

[1] Liu, Z. (Advisor), **Chen, X.***, Tian, F.*, Zhu, W., Hu, Z., Su, W., & Wang, Z. (2023). Dynamic effects of rough elements in fractures on coal permeability during different stages of methane extraction. *Gas Science and Engineering*, 119, 205118. DOI: 10.1016/j.jgsce.2023.205118.
[2] **Chen, X.**, Huang, S., Guan, T., Fu, H., & Pan, Z. (Accepted). An interpretable predictive model for liquid holdup of hilly-terrain oil-gas pipelines based on machine learning and Shapley additive explanations. *ACS Omega*
[3] **Chen, X.**, Huang, S., Guan, T., Fu, H., & Pan, Z. (Accepted). Prediction model of liquid holdup of multiphase pipelines based on PSO-BP algorithm. *Chemical Engineering (China)*.
[4] **Chen, X.***, Tang, Z., & Mo, Y. (2023). Design, Processing and Performance Testing of Extra-Large-Flow Centrifugal Model Pumps. *Journal of Physics: Conference Series*, 2610(1), 012022. DOI: 10.1088/1742-6596/2610/1/012022.
[5] Yang, G., **Chen, X.***, Hu, Y., Huang, S., & Xu, H. (2024). Analysis of vibration characteristics of cryogenic insulated cylinders based on ANSYS. *Journal of Physics: Conference Series*, 2860(1), 012016. DOI: 10.1088/1742-6596/2860/1/012016.
[6] Xie, H., Ou, C., **Chen, X.**, & Yang, G. (2023). Numerical analysis of the temperature rise at fast filling of hydrogen storage cylinder based on fluid-thermal-solid coupling. *Journal of Physics: Conference Series*, 2610(1), 012025. DOI: 10.1088/1742-6596/2610/1/012025.

PATENTS

[1] Huang, S. (Advisor), **Chen, X.**, Guan, T., & Fu, H. (2024). Prediction method for liquid holdup of hilly-terrain oil-gas pipelines based on SSA-BP neural network. (Chinese Patent No. 202410626009.7)
[2] Huang, S. (Advisor), **Chen, X.**, Hu, Y., & Yang, G. (2024). A finite element analysis method for vibration characteristics of vertical low temperature insulated cylinder system. (Chinese Patent No. 202411477988.0)

RESEARCH PROJECTS

AI-based predictive model for flow characteristic in hilly-terrain multiphase flow pipelines

Principal Investigator **|** *Advisor: Prof. Si Huang* *Jun. 2023 – Present*

- Simulated the oil-gas flow in a hilly-terrain multiphase pipeline under 18,900 different operating conditions using the multiphase flow simulation software OLGA
- Developed six AI-based prediction models, utilizing a dataset of simulation results
- Established an interpretable model integrating SHapley Additive exPlanations (SHAP) to reveal the global contributions and interactions of factors in the AI black box

Influence of fracture morphology on gas migration patterns in coal

Principal Investigator **|** *Advisor: Prof. Zhengdong Liu* *Mar. 2021 – Sep. 2023*

- Derived a new numerical model of rough-walled fractures that innovatively simplifies the fracture surface
- Proposed a concept of effective fracture aperture that varies nonlinearly with the CH_4 extraction process
- Constructed a CFD model of methane extraction using COMSOL Multiphysics, in which the governing equations for gas diffusion and seepage influenced by the new numerical model were imported
- Analyzed the dynamic effect of roughness, stagger degree and shape of rough elements on CH_4 migration

Simulation and testing of vibration characteristics of cryogenic insulated cylinders

Principal Investigator **|** *Advisor: Prof. Si Huang* *May 2024 – Present*

- Formulated a modal analysis model for the cryogenic insulated cylinder utilizing ANSYS
- Delved into the natural frequency trends of the cylinder across varying filling percentages

Fluid-thermal-solid coupling of carbon fiber full-wrapped hydrogen cylinders during fast inflating

Co-investigator **|** *Advisor: Prof. Si Huang* *Jan. 2023 – Jun. 2023*

- Established a fluid-thermal-solid coupled CFD model for fast inflating of carbon fiber full-wrapped hydrogen cylinders based on ANSYS
- Verified that the maximum stress point on the cylinder corresponds precisely to the vulnerability pinpointed by acoustic emission signal mapping

Hydraulic model design and test of large capacity vertical centrifugal pump

Co-investigator **|** *Advisor: Prof. Si Huang* *Sep. 2022 – Jan. 2023*

- Designed a model pump based on a prototype pump according to similarity principles
- Constructed physical models of centrifugal pumps utilizing CFTurbo
- Established the CFD model of centrifugal pumps using CFX

HONORS & AWARDS

South China University of Technology Graduate Scholarship (three times, RMB18,000) *Sep. 2022 – Jun. 2025*

Merit Undergraduate Thesis, Northeastern University (Top 2%) *Jun. 2022*

Likang Group Safety Engineering Scholarship (RMB 3,000) *Oct. 2021*

The National Undergraduate Mathematics Competitions, Third prize of national *Dec. 2020*

The National Undergraduate Mathematics Competitions, Second prize of provincial *Dec. 2020*

Northeastern University Merit Student Scholarship, Third prize (four times) *Sep. 2018 – Jun. 2022*

ACADEMIC CONFERENCES

The 4th International Conference on Fluid and Chemical Engineering, Wuhan, China *July 26 – 27 2024*

The 3th International Conference on Fluid and Chemical Engineering, Wuhan, China *July 21 – 22 2023*

SKILLS

Technology: ANSYS, COMSOL Multiphysics, OLGA, MATLAB, SpaceClaim

Language: Mandarin (Native), English (IELTS: 6.5), Cantonese (Basic)